

ARSALAN AMIR ALI

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EDUCATION

York University — Lassonde School of Engineering

Sep 2024 – Apr 2029 (Expected)

BSc Computer Science (Specialized Honours), Specialization in Artificial Intelligence

Relevant coursework: Data Structures · Operating Systems · Database Systems · Computer Organization · Theory of Computation · Advanced OOP · Probability · Linear Algebra

Awards: International Entrance Award for Academic Excellence · Lassonde Entrance Scholarship

Beaconhouse School System

Sep 2022 – Jul 2024

IB Diploma Programme, Computer Science · Habib University Meritorious Award (top 10 students) · 75% merit scholarship

TECHNICAL SKILLS

Languages: Python · Java · C · JavaScript · SQL

AI / ML: LLM APIs (Claude, OpenAI, Azure OpenAI, AWS Bedrock) · RAG · AI Agents (LangGraph) · MCP · LLM Evaluation · LLM Observability (Langfuse) · Prompt Engineering · Vector Search (pgvector) · PyTorch · Fine-Tuning (LoRA/QLoRA, Unsloth) · Triton GPU Kernels

Backend & Data: FastAPI · Node.js · Express · REST APIs · PostgreSQL · Supabase · Redis

Frontend & DevOps: Next.js · React · Docker · Git/GitHub · Linux · CI/CD (GitHub Actions) · Railway

Certifications: Anthropic Academy — Claude API · Model Context Protocol (Intro + Advanced)

EXPERIENCE & PROJECTS

ML Systems Engineer (Independent Project) — KernelForge

Jul 2026 – Present

Python · PyTorch · Triton · LoRA/QLoRA (Unsloth) · github.com/Arsalan-05/KernelForge

- Fine-tuning an LLM to generate and optimize Triton GPU kernels from PyTorch ops, scoped to LLM-inference patterns: quantized matmul, attention variants, KV-cache ops, and normalization fusion
- Built and GPU-validated an automated verification harness: correctness checks against PyTorch eager execution, speedup benchmarking, and sandboxed timeout isolation for untrusted generated code
- Building a verification-gated dataset and LoRA fine-tuning pipeline as a data-efficient, explainable alternative to broad-coverage systems (KernelLLM, Kevin-32B)

Founder & AI Software Engineer — FinSight AI

May 2026 – Present

Python · FastAPI · LangGraph · MCP · PostgreSQL + pgvector · Azure OpenAI · AWS Bedrock · Langfuse · Next.js · Docker · Railway · github.com/Arsalan-05/FinSight_AI

- Built solo and shipped to production (v2.0) a personal finance agent that grounds spending analysis in a RAG knowledge base over transaction history (chunking, embeddings, pgvector retrieval)
- Engineered stateful LangGraph agent workflows with persistent memory and MCP-connected tools, hardened with step and cost limits, retries, timeouts, fallbacks, and prompt-injection guardrails
- Built an automated eval suite that runs in CI (retrieval recall@k and answer-quality metrics) plus end-to-end tracing of every LLM and tool call, tracking latency, tokens, and cost per request
- Added model routing across Azure OpenAI and AWS Bedrock with response caching; deployed the full stack (Next.js frontend, FastAPI backend, PostgreSQL + pgvector) on Railway with Docker and GitHub Actions CI

Co-Founder & AI Software Engineer — RoomFinderAI

May 2025 – Present

Node.js · Express · Supabase · PostgreSQL · OpenAI · Cloudflare Workers AI · Stripe · Railway · roomfinderai.com

- Co-built an AI rental search, negotiation, and marketplace platform now live on web, iOS (App Store), and Android (Google Play): 60+ REST endpoints, 27 SQL migrations, and real-time chat via Supabase Realtime
- Engineered an LLM negotiation engine with an 8-phase conversation flow, landlord personality modeling, and a learning loop across 1,247 negotiations, reaching an 82% win rate on the top-performing template
- Built a true-cost calculator fusing 15+ external APIs (RentCast, Google Distance Matrix, GreatSchools, FEMA, BLS) and property-photo AI (Llama 3.2 Vision) that grades units A+ to C-
- Implemented the auth stack (email, Google OAuth, Apple Sign-In), Stripe payments, and Azure identity verification

INVOLVEMENT

York University: CS Hub · AWS Cloud Club · Data Science Club · AI & Data Science Club · YU Blueprint

Volunteer: AV Crew – Hackathon 3.0 · Event Coordinator – Sports Gala · Technical Crew – Drama Festival · Food Stall Coordinator – Flood Relief Carnival